

MATHEMATICS
Extension 2
Assessment task 3: 25/06/09

Time: 65 minutes.

SECTION 1.

Question 1.

For the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ find:

- a) The eccentricity. 2
- b) The co-ordinates of the vertices. 1
- c) The co-ordinates of the foci. 1
- d) The equation of the directrices. 1
- e) The equation of the asymptotes. 1
- f) Sketch the hyperbola. 2

Question 2.

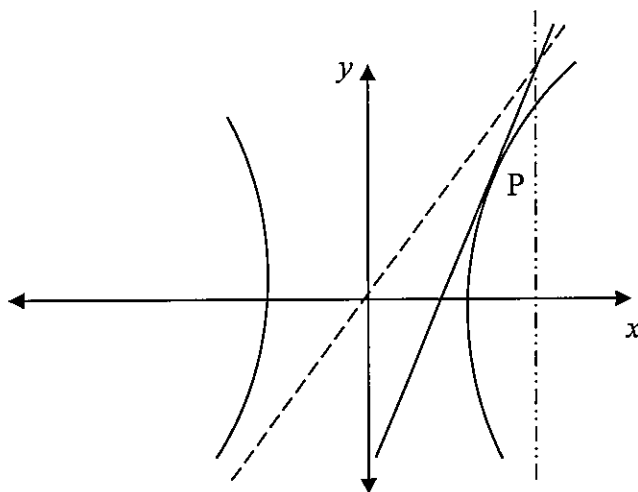
Show that the equation of the tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

at the point $P(a \cos \theta, b \sin \theta)$ is given by $\frac{\cos \theta x}{a} + \frac{\sin \theta y}{b} = 1$ 3

Question 3.

- a) Show that the equation of the tangent to the hyperbola $xy = c^2$
at the point $(cp, \frac{c}{p})$ is $x + p^2 y = 2cp$. 3
- b) If the tangents at the points P and Q meet at the point R (x_0, y_0)
prove that $pq = \frac{x_0}{y_0}$ and $p + q = \frac{2c}{y_0}$ 4

Question 4.



The point $P(a \sec \theta, b \tan \theta)$ on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ with focus, S , is such that the tangent at P , the latus rectum through S , and one asymptote are concurrent. Prove that SP is parallel to the other asymptote. (you may assume the equation of the tangent at P is given by $\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$).

4

SECTION 2.

Question 1

- a) Sketch the curve $y = \sin^{-1} x$, for $-1 \leq x \leq 1$ **1**
- d) By taking slices perpendicular to the axis of rotation use the method of slicing to find the volume of the solid generated by rotating the region bounded by the curve $y = \sin^{-1} x$, the ' x ' axis and the ordinate $x = 1$ about the ' y ' axis. **3**

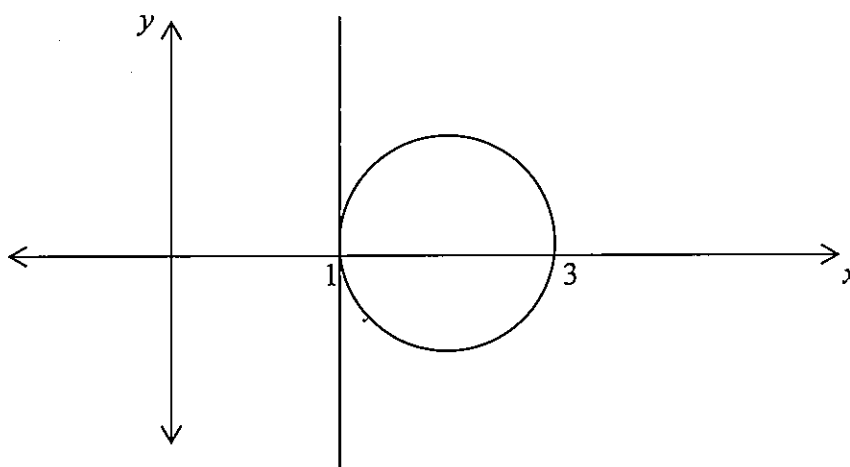
Question 2

The area between the curve $y = 8x - x^2$, the x axis and the line $x = 4$ is rotated about the line $x = 4$. Find the volume generated by using:

- a) slicing **3**
- b) cylindrical shells. **3**

Question 3.

a)

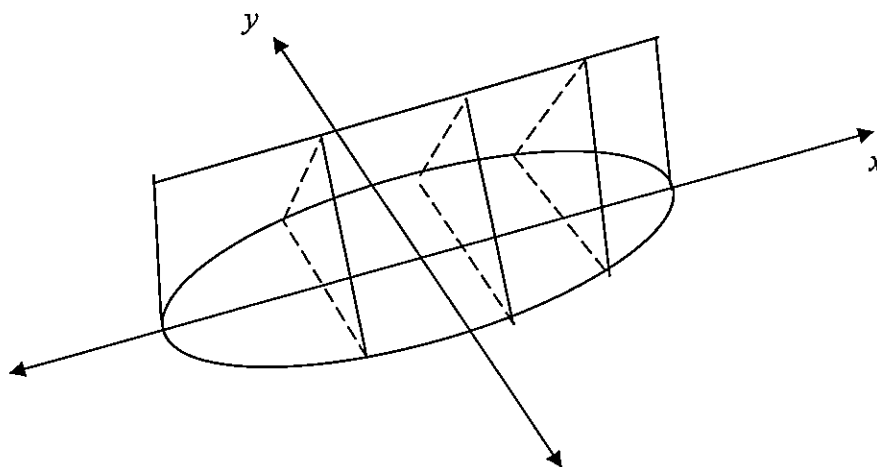


In the diagram above the circle $(x-2)^2 + y^2 = 1$ is drawn. The region bounded by the circle is rotated about the line $x = 1$. Use the method of cylindrical shells to show that the volume of the solid of revolution so formed is given by .

$$V = 4\pi \int_1^3 (x-1)\sqrt{1-(x-2)^2} dx \quad 3$$

ii) By using the substitution $x-2 = \sin u$, or otherwise, calculate the volume of the solid of revolution. 3

Question 4



A solid has a base in the form of an ellipse with major axis 10 units and minor axis 8 units. Find the volume if every section perpendicular to the major axis is an isosceles triangle with altitude 6 units.

3

EXTENSION 2. ASSESSMENT 3 SOLUTIONS (2009)

Ques 1) $\frac{x^2}{16} - \frac{y^2}{9} = 1$

a) $b^2 = a^2(e^2 - 1)$
 $9 = 16(e^2 - 1)$

$\frac{9}{16} = e^2 - 1$

$e^2 = \frac{25}{16}$

$e = \frac{5}{4}$

b) $(4, 0), (-4, 0)$

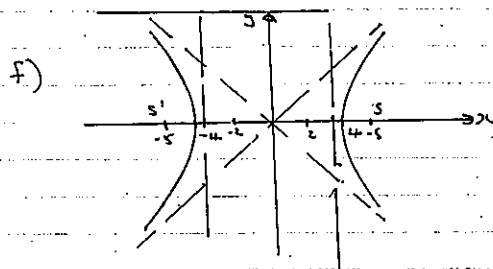
c) foci $(ae, 0), (-ae, 0)$
 $(5, 0), (-5, 0)$

d) $x = \frac{a}{e}, x = -\frac{a}{e}$

$x = \frac{16}{5}, x = -\frac{16}{5}$

e) $y = \frac{bx}{a}, y = -\frac{bx}{a}$

$y = \frac{3x}{4}, y = -\frac{3x}{4}$



Q2) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$\frac{2x}{a^2} + \frac{2y}{b^2} \cdot \frac{dy}{dx} = 0$

$\frac{2y}{b^2} \cdot \frac{dy}{dx} = -\frac{2x}{a^2}$

$\frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$

at $(a \cos \theta, b \sin \theta)$

$\frac{dy}{dx} = \frac{-b^2 a \cos \theta}{a^2 b \sin \theta}$

$= -\frac{b \cos \theta}{a \sin \theta}$

$\therefore$ eqn of the tangent

$y - b \sin \theta = \frac{-b \cos \theta}{a \sin \theta} (x - a \cos \theta)$

$a \sin \theta y - ab \sin^2 \theta = -b \cos \theta x + ab \cos^2 \theta$

$b \cos \theta x + a \sin \theta y = ab(\sin^2 \theta + \cos^2 \theta)$

$b \cos \theta x + a \sin \theta y = ab$

$\frac{\cos \theta x}{a} + \frac{\sin \theta y}{b} = 1$

Q3) a) $xy = c^2$

$x \cdot \frac{dy}{dx} + y = 0$

$\frac{dy}{dx} = -\frac{y}{x}$

at $(cp, \frac{c}{p}), \frac{dy}{dx} = \frac{-\frac{c}{p}}{cp}$

$= -\frac{1}{p^2}$

$\therefore$ equation of the tangent

$y - \frac{c}{p} = -\frac{1}{p^2}(x - cp)$

$p^2 y - cp = -x + cp$

$x + p^2 y = 2cp$

b) tangent at P: $x + p^2 y = 2cp$

tangent at Q: $x + q^2 y = 2cq$

Now (x_0, y_0) satisfies both equations

$\therefore x_0 + p^2 y_0 = 2cp \quad \dots (1)$

$x_0 + q^2 y_0 = 2cq \quad \dots (2)$

(1) - (2) $(p^2 - q^2) y_0 = 2cp - 2cq$

$(p+q)(p-q) y_0 = 2c(p-q)$

$y_0 = \frac{2c}{p+q}$

$p+q = \frac{2c}{y_0}$

(1) $xq^2: q^2 x_0 + p^2 q^2 y_0 = 2cpq^2 \quad \dots (3)$

(2) $xp^2: p^2 x_0 + p^2 q^2 y_0 = 2cq p^2 \quad \dots (4)$

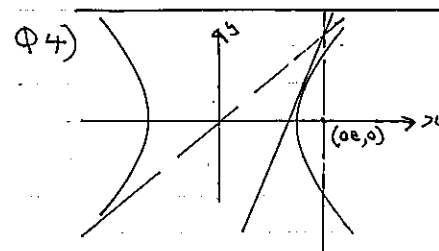
(4) - (3) $(p^2 - q^2) x_0 = 2cpq(p-q)$

$x_0 = \frac{2cpq(p-q)}{(p+q)(p-q)}$

$x_0 = \frac{2cpq}{p+q}$

$\therefore pq = \frac{x_0(p+q)}{2c}$

$= \frac{2c}{y_0} \quad (y_0 = \frac{2c}{p+q})$



eqn tangent at P: $\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$

eqn of asymptote: $y = \frac{bx}{a} \quad \dots (1)$

equation Latus rectum: $x = ae \quad \dots (2)$

Solving (1) and (2)

gives point of intersection

(ae, be)

Sub into the equation of the tangent

$\frac{ae \sec \theta}{a} - \frac{be \tan \theta}{b} = 1$

$e \sec \theta - e \tan \theta = 1$

$e(\sec \theta - \tan \theta) = 1$

$e = \frac{1}{\sec \theta - \tan \theta} \quad \dots (A)$

Now gradient SP

$= \frac{b \tan \theta}{a \sec \theta - ae}$

$= \frac{b \tan \theta}{a \sec \theta - a \left(\frac{1}{\sec \theta - \tan \theta} \right)} \quad \text{from (A)}$

$= \frac{b \tan \theta}{a \sec \theta - \frac{a}{\sec \theta - \tan \theta}}$

$= \frac{b \tan \theta (\sec \theta - \tan \theta)}{a \sec \theta (\sec \theta - \tan \theta) - a}$

$= \frac{b \tan \theta \sec \theta - b \tan^2 \theta}{a \sec^2 \theta - a \sec \theta \tan \theta - a}$

$= \frac{b \tan \theta \sec \theta - b(\sec^2 \theta - 1)}{a(\sec^2 \theta - \sec \theta \tan \theta - 1)}$

$= \frac{b(\tan \theta \sec \theta - \sec^2 \theta + 1)}{a(\sec^2 \theta - \sec \theta \tan \theta - 1)}$

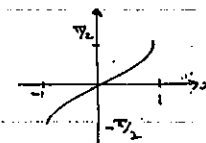
$= -\frac{b}{a}$

which is the gradient of the other asymptote.

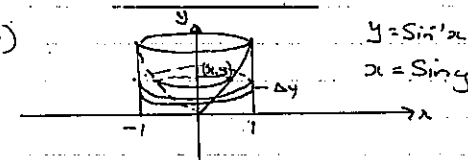
$\therefore SP \parallel$ to the other asymptote.

SECTION 2

Q1) a)



b)



Volume of Slice = $\pi(1^2 - x^2) \Delta y$

$V \approx \sum_{y=0}^{\pi/2} \pi(1 - x^2) \Delta y$

$V = \lim_{\Delta y \rightarrow 0} \sum_{y=0}^{\pi/2} \pi(1 - x^2) \Delta y$

$= \pi \int_0^{\pi/2} 1 - x^2 dy$

3.

$$\pi \int_0^{\pi/2} (1 - \sin^2 y) dy$$

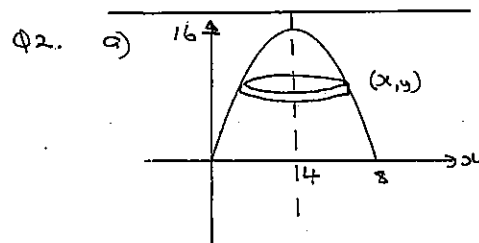
$$\pi \int_0^{\pi/2} \cos^2 y dy$$

$$= \frac{\pi}{2} \int_0^{\pi/2} \cos 2y + 1 dy$$

$$= \frac{\pi}{2} \left[\frac{1}{2} \sin 2y + y \right]_0^{\pi/2}$$

$$= \frac{\pi}{2} \left[(0 + \frac{\pi}{2}) - 0 \right]$$

$$= \frac{\pi^2}{4} \text{ cubic units.}$$



Volume of a Slice = $\pi(x-4)^2 \Delta y$

$$V \doteq \sum_{y=0}^{16} \pi(x-4)^2 \Delta y$$

$$V = \lim_{\Delta y \rightarrow 0} \sum_{y=0}^{16} \pi(x-4)^2 \Delta y$$

$$V = \pi \int_0^{16} (x-4)^2 dy$$

$$y = 8x - x^2$$

$$x^2 - 8x + y = 0$$

$$x = \frac{8 \pm \sqrt{64 - 4y}}{2}$$

$$= 4 \pm \sqrt{16 - y}$$

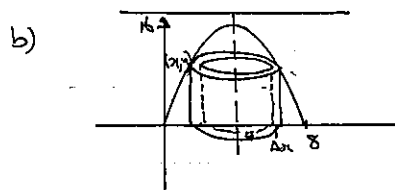
$$V = \pi \int_0^{16} (4 + \sqrt{16 - y} - 4)^2 dy$$

$$= \pi \int_0^{16} 16 - y dy$$

$$= \pi \left[16y - \frac{y^2}{2} \right]_0^{16}$$

$$\pi [(256 - 128) - 0]$$

$$= 128\pi \text{ cubic units}$$



Volume of a shell = $2\pi rh \cdot \Delta x$

$$= 2\pi(4-x)y \Delta x$$

$$V \doteq \sum_{x=-2}^2 2\pi(4-x)y \Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_{x=-2}^2 2\pi(4-x)y \Delta x$$

$$= 2\pi \int_{-2}^2 (4-x)y dx$$

$$= 2\pi \int_{-2}^2 (4-x)(4-x^2) dx$$

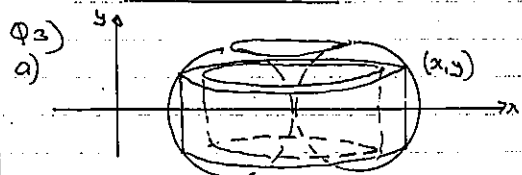
$$= 2\pi \int_{-2}^2 (x^3 - 12x^2 + 32x) dx$$

$$= 2\pi \left[\frac{x^4}{4} - 4x^3 + 16x^2 \right]_{-2}^2$$

$$= 2\pi [(64 - 256 + 256) - 0]$$

$$= 2\pi \times 64$$

$$= 128\pi \text{ cubic units.}$$



Volume of shell = $2\pi rh \cdot \Delta x$

$$= 2\pi(x-1)2y \Delta x$$

$$V \doteq \sum_{x=1}^3 2\pi(x-1)2y \Delta x$$

4.

$$V = \lim_{\Delta x \rightarrow 0} \sum_{x=1}^3 4\pi(x-1)y \Delta x$$

$$= 4\pi \int_1^3 (x-1)y dx$$

$$= 4\pi \int_1^3 (x-1)\sqrt{1-(x-2)^2} dx$$

$$\begin{cases} (x-2)^2 + y^2 = 1 \\ y^2 = 1 - (x-2)^2 \\ y = \sqrt{1 - (x-2)^2} \end{cases}$$

b) $V = 4\pi \int_1^3 (x-1)\sqrt{1-(x-2)^2} dx$

$$x-2 = \sin u \quad : x=1 \quad u = -\frac{\pi}{2}$$

$$x = \sin u + 2 \quad x=3 \quad u = \frac{\pi}{2}$$

$$\frac{dx}{du} = \cos u$$

$$dx = \cos u du$$

$$V = 4\pi \int_{-\pi/2}^{\pi/2} (\sin u + 1)\sqrt{1 - \sin^2 u} \cdot \cos u du$$

$$= 4\pi \int_{-\pi/2}^{\pi/2} (\sin u + 1)\sqrt{\cos^2 u} \cdot \cos u du$$

$$= 4\pi \int_{-\pi/2}^{\pi/2} (\sin u + 1)\cos^2 u du$$

$$= 4\pi \int_{-\pi/2}^{\pi/2} \sin u \cos^2 u + \cos^2 u du$$

$$= 4\pi \int_{-\pi/2}^{\pi/2} \sin u \cos^2 u + \frac{1}{2}(\cos 2u + 1) du$$

$$= 4\pi \int_{-\pi/2}^{\pi/2} \sin u \cos^2 u + \frac{1}{2}\cos 2u + \frac{1}{2} du$$

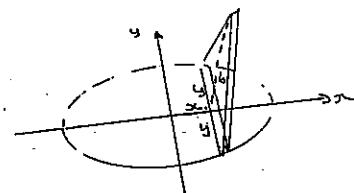
$$= 4\pi \left[-\frac{\cos^3 u}{3} + \frac{\sin 2u}{4} + \frac{u}{2} \right]_{-\pi/2}^{\pi/2}$$

$$= 4\pi \left[(0 + 0 + \frac{\pi}{4}) - (0 + 0 - \frac{\pi}{4}) \right]$$

$$= 4\pi \times \frac{\pi}{2}$$

$$= 2\pi^2 \text{ cubic units.}$$

Q4)



equation of the ellipse: $\frac{x^2}{25} + \frac{y^2}{16} = 1$

Volume of a Slice = $\frac{1}{2} \times 2y \times 6 \cdot \Delta x$

$$V \doteq \sum_{x=-5}^5 6y \Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_{x=-5}^5 6y \Delta x$$

$$V = 6 \int_{-5}^5 y dx$$

$$\begin{cases} \frac{x^2}{25} + \frac{y^2}{16} = 1 \\ y^2 = 16(1 - \frac{x^2}{25}) \\ = 16(\frac{25-x^2}{25}) \\ y = \frac{4}{5}\sqrt{25-x^2} \end{cases}$$

$$V = \frac{24}{5} \int_{-5}^5 \sqrt{25-x^2} dx$$

$$= \frac{24}{5} \times (\frac{1}{2} \pi r^2)$$

$$= \frac{12}{5} \pi \times 5^2$$

$$= 60\pi \text{ cubic units.}$$